

WHERE HAS THE MONEY GONE?

U.S. Labor Market & Industry Distribution Analysis

Analytical & Research Framework

American Economic Data Repository (AEDR)

Part of the American Dream Affordability research program

Where Has the Money Gone Project Summary

This document defines the analytical scope, workflow, and guiding principles for Where Has the Money Gone?, the first major analytical project built from the American Economic Data Repository (AEDR). The project examines how employment, wages, and economic gains have been distributed across the U.S. labor market over time.

The baseline analysis is intentionally broad. It begins at the national 2-digit NAICS sector level and uses that level to identify long-run structural patterns in employment, compensation, and the distribution of the national wage pool. More detailed industry, ownership, occupational, and geographic analyses are treated as optional drill-downs when the broad results identify a question that warrants deeper investigation.

This structure keeps the initial project interpretable and reproducible while preserving a clear path for progressively more detailed research.

AEDR and the Larger Research Program

American Economic Data Repository (AEDR)

AEDR is the underlying public economic research repository supporting this project and future analyses. Its purpose is to preserve, validate, organize, and document economic data from major public sources so that analytical projects can be built from a consistent and reproducible foundation rather than from one-off data extracts.

AEDR includes or is designed to include labor and income data, housing data, demographic data, geographic reference data, and macroeconomic indicators. Major source families include BLS QCEW and OEWS, Census/ACS, FHFA housing data, FRED and other Federal Reserve series, BEA data, and supporting geographic and classification reference files.

The repository is not itself the final research conclusion. It is the data infrastructure from which question-specific analytical datasets are constructed.

American Dream Affordability Research Program

The larger research program asks how the economic affordability of achieving a traditional American standard of living has changed over time and across the United States. Rather than attempting to answer that question with a single model or index at the outset, the work is divided into three related analytical projects that follow a logical research sequence.

Project	Core Question	Primary Role
Where Has the Money Gone?	How have jobs, wages, and labor-market gains been distributed across the U.S. economy?	Establish the economic-opportunity side of affordability.
Housing Affordability	How have housing costs changed relative to incomes and broader economic conditions?	Establish the housing-cost side of affordability.
Where Is the American Dream Most Affordable?	Where do employment opportunity, earnings, housing affordability, and broader conditions combine most favorably?	Integrate the preceding analyses and support development of the American Dream Affordability Index.

The American Dream Affordability Index (ADAI) is therefore an eventual output of the broader research program, not a predetermined formula. Its construction should follow the underlying labor, housing,

geographic, and macroeconomic analysis so that variables, transformations, weights, and assumptions are informed by the evidence rather than selected in advance.

1. Core Research Question

Where have the gains in wages, employment, and economic opportunity gone within the U.S. labor market?

The phrase “where has the money gone?” is a broad research problem rather than a literal target variable. The project translates that question into measurable descriptive tasks involving the distribution and change of employment, wages, pay, and industry shares through time.

At the baseline level, the project asks:

- Which broad sectors experienced the strongest and weakest employment growth?
- Which sectors captured larger or smaller shares of national employment over time?
- Which sectors captured larger or smaller shares of the national wage pool?
- How did nominal and inflation-adjusted average pay change across sectors?
- Did changes in wage share occur alongside similar changes in employment share?
- Which sectors experienced the largest divergence between their share of workers and their share of wages?
- How concentrated or redistributed were employment and wages across sectors from year to year?
- Which periods exhibit unusual structural movement that warrants additional investigation?

2. Baseline Unit of Analysis

National 2-Digit NAICS Sectors

The broad analytical baseline is the national 2-digit North American Industry Classification System (NAICS) sector level. This provides a consistent, interpretable view of the major parts of the U.S. economy before introducing finer industry detail.

The completed analytical notebook validates 21 national sector-level classifications across 1990–2025, with 20 substantive classified sectors represented throughout the period and NAICS 99 — Unclassified appearing beginning in 2001. Combined sector identifiers such as 31–33 Manufacturing, 44–45 Retail Trade, and 48–49 Transportation and Warehousing are treated as valid sector-level classifications.

The analytical workflow begins at this broad level because it allows structural patterns to be identified without prematurely fragmenting the analysis into thousands of lower-level industry, ownership, or geographic combinations.

Observation Grain and Aggregation Discipline

Every analytical dataset must have an explicitly defined observation level. QCEW contains multiple geographic, ownership, industry, and aggregation levels, so records from different grains cannot be treated as directly comparable simply because they appear in the same source.

The notebook therefore validates the relevant national sector structure, ownership coverage, observation uniqueness, and aggregation behavior before constructing the broad analytical series. Aggregate ownership categories are not double-counted alongside their component categories.

3. Primary Data and Analytical Coverage

QCEW

The baseline labor-market analysis uses the AEDR QCEW analytical Parquet covering 1990–2025. QCEW provides the long-run industry measures required for the core project, including employment, establishments, total annual wages, average weekly wage, average annual pay, ownership, geography, and industry classifications.

Inflation and Real Pay

Nominal pay alone cannot establish whether workers gained purchasing power. The analysis therefore incorporates CPI-based inflation adjustment where required to compare compensation through time in real terms.

Historical Comparability

The workflow preserves documented historical source differences instead of manufacturing a perfectly balanced panel. In particular, pre-2001 QCEW industry data were originally classified under SIC and subsequently reconstructed by BLS into historical NAICS-based data. The completed notebook retains those observations while explicitly flagging the additional comparability limitation.

Likewise, absent source-derived industry or ownership cells are not automatically converted to zero. Missing, suppressed, unavailable, and structurally nonexistent observations are treated as conceptually different conditions.

4. Analytical Workflow

Stage 1 — Data Understanding and Validation

Before substantive analysis, the analytical dataset is profiled and validated to establish what each observation represents and whether the intended analytical slice behaves consistently with the source data and AEDR engineering decisions.

- Confirm year and sector coverage.
- Validate 2-digit NAICS mappings and combined sector identifiers.
- Inspect ownership coverage without assuming every ownership × industry combination must exist.
- Validate observation uniqueness at the intended grain.
- Confirm aggregation behavior and avoid double counting published totals and component records.
- Review nulls, suppression, source-cell absence, extreme values, and structural breaks.
- Document classification and historical-comparability boundaries before interpreting results.

Stage 2 — Establish the National Structural Baseline

The first substantive analysis establishes the broad distribution of U.S. employment and wages across 2-digit NAICS sectors. The objective is to understand the size, direction, and long-run movement of the major sectors before examining more detailed classifications.

- National employment by sector.
- Total annual wages by sector.
- Average annual pay by sector.

- Nominal and real pay growth.
- Employment share of national totals.
- Wage share of national totals.
- Long-run change in those shares.

Stage 3 — Real Pay Growth Across Industries

Inflation-adjusted average annual pay is used to evaluate whether compensation gains translated into improved purchasing power. The completed notebook finds positive full-period real average annual pay growth across every comparable broad sector, but the magnitude of improvement is highly uneven.

The analytical emphasis is therefore not simply whether average pay rose faster than inflation, but how differently those gains were distributed across sectors.

Stage 4 — Employment and Wage Distribution

Sector employment share and wage share are analyzed together to distinguish growth in the number of workers from growth in the share of compensation captured by a sector. This allows the project to identify structural changes that absolute wage or employment totals alone can obscure.

The analysis includes long-run sector share changes as well as year-over-year movements in the national distribution of employment and wages.

Stage 5 — Redistribution and Divergence

The completed notebook develops descriptive measures of annual employment redistribution and annual wage redistribution across broad sectors. These measures quantify the magnitude of year-to-year changes in sector shares without implying that workers or dollars literally moved from one industry to another.

A wage-versus-employment redistribution gap compares the magnitude of those two distributional movements. Industry-level wage-share versus employment-share divergence then identifies the specific sector-year observations in which the relative distribution of wages changed substantially differently from the relative distribution of employment.

These are descriptive measures. They identify when and where distributions changed; they do not independently establish why those changes occurred.

Stage 6 — Interpret Findings Conservatively

The project separates measurement from causal explanation. Correlations, share changes, and divergence patterns are used to identify economically interesting relationships and periods, but they are not treated as proof of causation.

Unexpected results are validated against the underlying source-derived values and documented limitations before being interpreted. Historical reconstruction, classification changes, publication structure, and aggregation choices are considered potential sources of comparability issues rather than silently ignored.

5. Completed Baseline Analysis — What It Establishes

The completed notebook demonstrates the intended broad workflow and establishes a reproducible national sector-level baseline for 1990–2025.

- Validated national 2-digit NAICS sector structure and historical coverage.

- Validated component ownership coverage and the distinction between absent cells and suppressed observations.
- Validated analytical grain and aggregation behavior before calculating derived measures.
- Measured long-run employment-share and wage-share changes across broad sectors.
- Converted average annual pay to real terms for long-run purchasing-power comparison.
- Compared employment-share change with real-pay growth.
- Measured annual employment redistribution and annual wage redistribution.
- Compared wage redistribution with employment redistribution.
- Identified the largest sector-year wage-share versus employment-share divergences.
- Documented the additional comparability limitation associated with pre-2001 reconstructed NAICS observations.

The broad substantive conclusion is that long-run real average pay improved across the comparable sector set, but the gains were highly uneven. The completed analysis therefore frames “where has the money gone?” primarily as a question of the uneven distribution of real-pay growth and changes in the relative wage and employment position of major industries, rather than as evidence of a universal decline in inflation-adjusted average pay.

6. Optional Analytical Drill-Downs

The broad 2-digit sector analysis is the starting point, not the maximum level of detail supported by AEDR. Once a pattern is identified, the research can drill down selectively. A drill-down should be motivated by a finding or research question rather than performed simply because more granular data are available.

Finer NAICS Industry Analysis

Move from sectors into subsectors, industry groups, industries, or national industries to determine whether a broad sector result is widely distributed or driven by a smaller group of industries.

Ownership Analysis

Compare private, federal, state, and local government employment where the source structure supports meaningful comparisons. Ownership × industry analysis can be used when a broad sector pattern appears to differ between public and private employment.

Geographic Analysis

Introduce state, CBSA, or county geography to test whether national industry patterns are geographically concentrated or whether the same industry produced materially different wage and employment outcomes in different labor markets.

Occupational Analysis — OEWS

OEWS can extend the analysis from industries to occupations. This is particularly useful when the research needs to distinguish between an industry becoming more highly compensated in aggregate and broad groups of workers within that industry experiencing stronger wage outcomes.

Diagnostic and Explanatory Analysis

Additional AEDR, FRED, BLS, and other macroeconomic variables can be introduced after a descriptive pattern has been established. Potential explanatory variables may include unemployment, labor-force

participation, payroll growth, inflation, interest rates, productivity where available, and broader macroeconomic conditions.

If supervised modeling is used, the target variable must be explicitly defined from the phenomenon identified in the descriptive analysis. The model type follows from the research problem; machine learning is not treated as an objective by itself.

Predictive or Prescriptive Extensions

Prediction should be attempted only when a well-defined target, appropriate historical coverage, and time-aware validation strategy exist. Prescriptive conclusions require an additional decision context and should not be added merely to label the project as prescriptive analytics.

7. Relationship to the Housing Affordability Project

Where Has the Money Gone? establishes the economic-opportunity side of the larger American Dream Affordability research program: how the ability to earn and the distribution of labor-market gains have changed.

The next major project examines housing affordability. It will evaluate housing costs relative to incomes and economic conditions using appropriate AEDR housing, household-income, interest-rate, and geographic data. Measures may include home values, rents, housing-price appreciation, income relationships, and financing conditions.

Keeping the labor-market and housing analyses initially separate prevents the final affordability framework from being defined before its major components are understood.

8. Relationship to “Where Is the American Dream Most Affordable?”

The third major analytical project will integrate the economic-opportunity findings from the labor-market analysis with the housing-affordability findings and relevant geographic and macroeconomic conditions.

The central question becomes: where do earnings, employment opportunity, housing affordability, and broader economic conditions combine to provide the strongest relative opportunity to achieve the standard of living being studied?

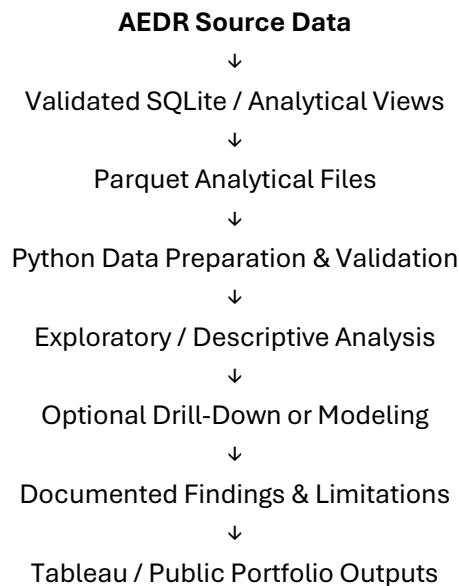
This stage is expected to make greater use of state, county, and metropolitan geography and will provide the empirical foundation for an eventual American Dream Affordability Index.

9. Modeling and Validation Principles

- Research questions come before algorithms.
- The observation grain must be explicit before analysis or modeling.
- Source differences and historical availability are preserved rather than hidden through artificial balancing or inappropriate imputation.
- Nominal and real measures are distinguished when purchasing power matters.
- Descriptive findings are established before causal or predictive claims are attempted.
- Temporal data require time-aware training and validation when predictive modeling is used.
- Simple, interpretable baselines should precede more complex models.
- Statistical performance must be accompanied by economic and domain validation.

- Robustness and sensitivity analysis are required before treating a narrow specification as a general economic finding.
- All transformations, exclusions, assumptions, and analytical decisions should be documented for reproducibility.

10. Reproducible Analytical Pipeline



11. Guiding Principle

The model is not the project.

The project is the disciplined use of economic data to understand how opportunity, costs, and affordability have changed, and ultimately where Americans can achieve the strongest standard of living relative to what they earn.

12. Current Scope Statement

The current Where Has the Money Gone? project is a national, long-run labor-market analysis centered on broad 2-digit NAICS sectors. Its completed baseline examines 1990–2025 QCEW employment and wage distributions, incorporates inflation-adjusted pay comparisons, and evaluates how the relative position of major industries changed through time.

Finer industry classifications, ownership, geography, OEWS occupations, diagnostic modeling, prediction, and prescriptive analysis remain supported extensions of the framework. They are not required components of the baseline project and should be introduced only when they materially improve the answer to a defined analytical question.

This scope preserves the broad purpose of the research while maintaining a clear analytical sequence: understand the labor market first, understand housing affordability second, and then integrate those findings to evaluate where the American Dream is most affordable.

Source and Method Context

This analytical framework is grounded in the AEDR QCEW analytical workflow and the broader American Dream Affordability data-science and modeling framework. Public-source documentation, including BLS QCEW classification, aggregation, and historical-comparability guidance, forms part of the project's methodological foundation and supports the interpretation, validation, and reproducibility of the analysis.